LESSON 12.4
FACTORING EXPRESSIONS

LESSON INTRODUCTION

MA.8.AR.1.3 Rewrite the sum of two algebraic expressions having a common monomial factor as a common factor multiplied by the sum of two algebraic expressions.

LEARNING TARGETS

- I can rewrite an algebraic expression with a common factor using the distributive property.

BENCHMARK COHERENCE

BUILDING ON

MA.7.AR.1.2 Determine whether two linear expressions are equivalent.

WORKING TOWARD

MA.912.AR.1.7 Rewrite a polynomial expression as a product of polynomials over the real number system.

ESSENTIAL QUESTIONS

- How do I rewrite an algebraic expression using a common factor?
- How can I use the distributive property to factor an expression?

LESSON NARRATIVE

This lesson activates prior knowledge of common factors and the Greatest Common Factor from prior grades to use when factoring expressions. Students explore factoring expressions that have a common factor using area models. The area model representation is used to connect the concept of using the distributive property to factor algebraic expressions that have a common factor.

COMPONENT SUMMARY

12.4.1 WARM-UP *(~5 MIN.)*

Complete a number puzzle

12.4.2 REFRESHER *(5-10 MIN.)*

Find common factors and identify Greatest Common Factor

12.4.3 ACTIVITY *(5-10 MIN.)*

Match equivalent numeric expressions

12.4.4 EXPLORATION *(15 MIN.)*

Use common factors to rewrite algebraic expressions

12.4.5 BRING IT TOGETHER *(10-15 MIN.)*

Factor expressions

12.4.6 TRY IT *(5-10 MIN.)*

Rewrite expressions in factored form

12.4.7 WRAP-UP *(~5 MIN.)*

Explain how to rewrite an algebraic expression in factored form

RESOURCES

GLOSSARY

Key Terms:

- Common Factor
- Greatest Common Factor

Additional Terms:

- N/A

MATERIALS

- Cards for Card Sort

ADDITIONAL PREPARATION

- N/A

TEACHER TIPS

COMMON MISCONCEPTIONS

- Students may struggle to rewrite expressions when negatives are involved. For example, 12.4.4 Exploration expanding -(2x-3y) students may not distribute the negative sign to both terms inside the parentheses.

THINGS TO CONSIDER

- Students studied the skill of finding common factors and the Greatest Common Factor (GCF) in 6th grade. GCF is not explicitly taught in 7th grade.
- If time is an issue, then consider assigning 12.4.6 Try It for homework.

LITERACY AND LANGUAGE ROUTINES

Take Turns

12.4.4 Exploration students can take turns writing an equivalent expression for one of the area models. One partner can complete the model and explain their reasoning while the other listens. If the partner disagrees, they work to resolve the discrepancy before moving to the next problem and switching roles.

MATHEMATICAL THINKING AND REASONING (MTR) STANDARD

MA.K12.MTR.2.1 Multiple Representations

12.4.4 Exploration provides students with collaborative, exploratory activity that allows students to conceptualize using a common factor to rewrite algebraic expressions. Students will build on their understanding of the concept through completing area models using common factors. Students will connect the area model representation to the concept of using the distributive method to factor algebraic expressions.

DIFFERENTIATE INSTRUCTION

In 12.4.6 Try It consider your students when determining whether students are working independently or with a partner. If students struggled with previous activities, consider allowing students to work in pairs for 12.4.6 Try It. If working with a partner, consider assigning specific roles and asking them to switch roles for rewriting each expression. For example, one student can verbally explain to their partner the steps to take to rewrite the expression. After the partner takes the exact steps explained, both students assess the expression to agree if it is equivalent. For the following expression, students would switch roles. Now, the student who was previously explaining the steps will be following the steps detailed by their partner. At some point in 12.4.6 Try It, have students attempt factoring expressions on their own and comparing answers with their partner. This will allow for a gradual release of support and attempt to build students' confidence in factoring.

12.4.1 WARM-UP: NUMBER PUZZLE

DIFFERENTIATE INSTRUCTION

Consider the needs of your students to determine if 12.4.1 Warm-Up would be more effective if done in pairs.

12.4.1 Warm-Up: Number Puzzle

1. Solve the puzzle by filling in each blank cell. The values in the blue box represent the products of the values at the end of each row and column.

49	84	7
21	36	3
7	12	

12.4.2 Refresher: Common Factors

In multiplication expressions, the numbers being multiplied are called factors and the result is called the product.

Numbers can have **common factors**.

A **common factor** of two numbers is a factor that they both share.

For example, $1, 3, 5, 9, 15$ and 45 are factors of 45, and $1, 2, 3, 4, 5, 6, 10, 12, 15, 20, 30,$ and 60 are factors of 60.

The common factors 45 and 60 are $1, 3, 5$, and 15.

The ***greatest common factor*** (GCF) of two or more whole numbers is the largest whole number that evenly divides the given whole numbers.

For example, 15 is the greatest common factor for 45 and 60.

1. For the number pairs in the table, list the common factors. Then, identify the GCF.

12 and 42	10 and 35
Common factors: $1, 2, 3, 6$ GCF $= 6$	Common factors: $1, 5$ GCF $= 5$

Common factors can be used to rewrite numeric expressions.

2. Consider the expression $24 + 32$.

   a. List all common factors of 24 and 32.
   $1, 2, 4, 8$

   b. Complete the steps to rewrite the sum of 24 and 32 using their GCF.
   $24 + 32$

   $(8 \times 3) + (8 \times 4)$

   $8(3 + 4)$

   c. Choose another common factor of 24 and 32 to write a different equivalent expression to $24 + 32$.
   Answers will vary based on which factor chosen.
   $24 + 32$

   $(4 \times 6) + (4 \times 8)$

   $4(6 + 8)$

12.4.2 REFRESHER: COMMON FACTORS

TEACHER MOVES

Students learned about GCF in 6th grade. As you are reviewing this concept with the class, ensure students have a firm understanding of how to find common factors and GCF. This is foundational for being able to utilize the distributive property to rewrite expressions in this lesson. If you notice students struggling, consider pulling small a small group for remediation during 12.4.3 Activity.

ENRICHMENT

- How can we find the common factors of 24 and 32?
- Is there more than one method to do this?
- What is the GFC of 24 and 32? How do you know?

TEACHER MOVES

Consider having some students display their answer to question 2c and explain the process they used to the class.

12.4.3 ACTIVITY: CARD SORT

TEACHER MOVES

Listen to students discuss their thinking and processes for identifying which expressions are equivalent. If some student pairs finish before others, ask those students to compare matches with another pair of partners and discuss the strategies used. As part of their discussion, have them determine how they verified the expressions were equivalent.

12.4.3 Activity: Card Sort

Six numeric expressions are shown below. Work with your partner to find at least one equivalent expression for each. Expressions may have one, two, or three matches.

Record the letter(s) of the equivalent expressions in the space provided under each expression below.

1. $90 - 40$ E, J	2. $18 + 66$ G, I	3. $20 - 80$ F, K, L
4. $-30 + 45$ A, C	5. $49 + 21$ D	6. $72 + 54$ B, H

12.4.4 Exploration: Equivalent Forms of Algebraic Expressions

Common factors can also be used to rewrite algebraic expressions.

1. Use your understanding of common factors to complete the area models shown.

	5	$-2y$
-3	-15	$6y$

	a	-6
5	$5a$	-30

	$3a$	$-b$
2	$6a$	$-2b$

	$4z$	-1
-4	$-16z$	4

	5	$-x$
6	30	$-6x$

	$2x$	-7
-4	$-8x$	28

Answers will vary for this model based on which factor was chosen.

12.4.4 EXPLORATION: EQUIVALENT FORMS OF ALGEBRAIC EXPRESSIONS

TEACHER MOVES

Take Turns

Consider arranging students in groups of 2 to complete question 1. Have students take turns in completing the area model and explain their thinking (or process) before the next student has a turn. For the last two area models, students can do those independently then take turns discussing how they completed each model.

TEACHER MOVES

Students may make errors during 12.4.4 Exploration, including forgetting to bring the sign with the term or multiplying and dividing with negatives incorrectly. Allow the opportunity for them to correct their own thinking as they work through the area models. Part of what students may realize by the end of 12.4.4 Exploration as they move into 12.4.5 Bring It Together is that using the area model can be a helpful visual when factoring. Do not be afraid to let the mistakes happen, only to discuss them during the transition to the next component.

12.4.4 EXPLORATION: EQUIVALENT FORMS OF ALGEBRAIC EXPRESSIONS (CONTINUED)

TEACHER MOVES

Observe and listen as students compare their answers and thinking with their partners. Make note of student ideas that you can ask them to share in the whole-group 12.4.5 Bring It Together.

An area model can be used to show the relationship between equivalent forms of algebraic expressions.

In the area models in the previous problem, the factored form of an expressions is the $width \times length$ of the larger rectangle. An equivalent form can be represented by the sum of the areas of each smaller rectangle.

2. Complete the table using the area models from the previous problem.

Factored Form	Equivalent Expression
$-3(5-2y)$	$-15+6y$
$5(a-6)$	$5a-30$
$2(3a-b)$	$6a-2b$
$-4(4z-1)$	$-16z+4$
$6(5-x)$	$30-6x$
$-4(2x-7)$	$-8x+28$

3. Compare your work with your partner. Discuss any differences and make any corrections as needed.

TEACHER MOVES

Students may make errors during question 4, including forgetting to multiply the term outside to both terms inside the parentheses, or to distribute the negative when multiplying. Allow the opportunity for them to correct their own thinking as they work through the question with their partner.

DIFFERENTIATE INSTRUCTION

Consider allowing students to share verbally with another classmate (and some share whole-group) their explanation to question 4b as opposed to making them write an explanation. Also, consider allowing students who prefer to share through using a representation (such as an area model or drawing of their explanation) to do so.

4. Work with your partner to complete the following.

   a. Without using area models, complete the table.

Factored Form	Equivalent Expression
$-4(2w-5z)$	$-8w+20z$
$-(2x-3y)$	$-2x+3y$
$10(2x-y)$	$20x-10y$
$4(a-11)$	$4a-44$
$5(2a-5b)$	$10a-25b$
$-2(3y-z)$	$-6y+2z$

   b. Explain how you determined the factored form of the expression $10a-25b$.
   Answers will vary. I found a common factor between $10a$ and $-25b$, 5. Then, I divided both terms by 5 to find what to write in the parentheses.

5. For the expression $20x - 10y$, Pao and Camiya wrote different factored forms of the expression.

Pao's Expression	Camiya's Expression
$5(4x - 2y)$	$10(2x - y)$

a. Explain who wrote a correct equivalent expression in factored form.
Both wrote a correct equivalent expression; they just factored out different common factors.

b. What is another equivalent expression using a different common factor?
$2(10x - 5y)$

c. Which expression is equivalent to $20x - 10y$ using the GCF of $20x$ and $-10y$?
$10(2x - y)$

12.4.4 EXPLORATION: EQUIVALENT FORMS OF ALGEBRAIC EXPRESSIONS (CONTINUED)

DIFFERENTIATE INSTRUCTION

For additional scaffolding, consider asking students before starting question 5a to determine the Greatest Common Factor of the expression 20x-10y.

12.4.5 BRING IT TOGETHER: FACTORING EXPRESSIONS

TEACHER MOVES

Review student responses to question 1 whole-group, so that students understand there are multiple answers depending on the factor chosen. Alternately, as students complete question 1, have them select a different factor and see how their answers change.

12.4.5 Bring it Together: Factoring Expressions

Properties of operations can be used in different ways to create equivalent expressions.

The distributive property can be used to expand an expression. For example, the expression $3(x+5)$ can be rewritten as $3(x)+3(5)$ or $3x+15$.

The distributive property can also be used to factor an expression. For example, the expression $8x+12$ can be rewritten as $2(4x+6)$ or $4(2x+3)$.

Equivalent expressions can also be generated using the commutative property of addition or multiplication.

$$(4x+6)2 \qquad 4(3+2x)$$

1. Complete the following to factor the expression $12d-36$.
   Answers can vary based on which factor is chosen.

   a. Choose a common factor of the terms $12d$ and -36.

   ○ 2 ○ 3 ○ 4 ○ 6 ● 12

   b. What is the result of dividing $12d$ by the common factor selected?
   d

   c. What is the result of dividing -36 by the common factor selected?
   -3

   d. Use your answers to Parts A-C to write an expression equivalent to $12d-36$ in factored form.

   $$12d - 36 = 12(d + -3)$$

e. Choose a different common factor from Part A, and use it to write a different equivalent expression.
Answers can vary depending on which factor is chosen.
$12d - 36 = 3(4d + -12)$

2. Several expressions are shown.

$8(3+m)$ *☺	$2(8m+12)$	$4(2m+6)$
$(2m+8)3$	$(1+3)8m$	$(12+4m)2$ ☺

a. Circle the expressions equivalent to $8m + 24$.

b. Put an asterisk (*) next to the equivalent expression that has the GCF of $8m$ and 24 as one of its factors.

c. Put a smiley face (☺) next to an expression where more than one property of operations can be used to show it is equivalent to $8m + 24$.

12.4.5 BRING IT TOGETHER: FACTORING EXPRESSIONS (CONTINUED)

ENRICHMENT

How did you decide which expressions in question 2 were equivalent to 8m+24?

QUESTIONING

For the expression with both an asterisk and a smiley face, identify the properties of operations that can be used to show it is equivalent to 8m+24.

12.4.6 TRY IT: FACTORING EXPRESSIONS

TEACHER MOVES

If time is an issue, then consider assigning 12.4.6 Try It for homework.

12.4.6 Try It: Factoring Expressions

1. Rewrite each expression in an equivalent factored form.
   a. $-15z + 20 = -5(3z - 4)$
   b. $4x - 32 = 4(x - 8)$ or $2(2x - 16)$
   c. $-27 - 12d = -3\ (9 + 4d)$
   d. $35f + 28g = 7(5f + 4g)$

2. Fill in the blanks of the equation to make it true.

$$75a + 25b = 25(3a + b)$$

12.4.7 Wrap-Up: Cool Down

Janelle missed math class today. She understands using the distributive property to expand an expression, but she doesn't know how to rewrite the sum of two terms with a common factor in factored form.

1. Explain to Janelle how to rewrite an expression like $36a - 16$ in an equivalent factored form.
   Answers will vary. First, you need to find a common factor between the two terms, like 4. *Then, you need to divide each term by that factor, which would give you* $9a - 4$. *The equivalent factored form is the product of the two factors:* $4(9a - 4)$.

12.4.7 WRAP-UP: COOL DOWN

DIFFERENTIATE INSTRUCTION

Consider allowing students to choose how they want to explain this process. For example, they could make a poster, a video, show steps and area models, etc.